Pelvic Viscera, Part 1

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Block: Endo/Uro/Repro

Thread: Anatomy

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Textbook

Moore's Clinically Oriented Anatomy, 9th ed; Go To: Chapter 6: Pelvis and Perineum

Learning Objectives

By exam time, you should be able to do the following:

1. Describe the branches of the internal iliac artery and the structures they Discuss the vasculature supply to the pelvis and perineum, including relationships and anastomoses with the systemic and portal systems
2. Discuss the somatic and autonomic innervations of the structures in the pelvis and perineum, including consequences of functional loss
3. Describe the general lymphatic drainage of the pelvis and perineum
4. Discuss the structural and functional anatomy of the pelvic portions of the urinary and gastrointestinal systems, including their anatomical relationships to abdominopelvic and perineal structures.

Study Questions

1. A 44-year-old patient presents with rectal bleeding. She describes the blood as bright red. She does not experience pain in the anal region. Further examination reveals internal hemorrhoids. You recall that internal hemorrhoids stimulate sensory fibers of which of the following nerves?

    A. Inferior gluteal

    B. Inferior hypogastric plexus

    C. Inferior rectal

    D. Obturator

    E. Posterior femoral cutaneous

2. A 42-year-oldpatient was diagnosed with ovarian cancer. The cancerous ovary was surgically removed.

    Question A: During the surgery, do you recall which nerve is vulnerable to injury?

    Question B: During the surgery, you need to ligate the ovarian vessels. Which structure might be at risk during this procedure?

3. Based on the branches of the internal iliac artery, which body regions receive arterial supply via internal iliac artery?

4. At which structures can we see the urinary system and reproductive system merge?

5. You are performing a rectal examination on your 72-year-old patient. Based on the pelvic anatomy you learned at medical school, recall what structures you can palpate in patients?

A. Neurovascular structures of the pelvis

1. Pelvic arteries
    a. Internal iliac arteries are the principal arteries supplying the pelvic viscera.
    b. Median sacral artery, superior rectal artery and ovarian arteries also provide arterial supply to the structures in the pelvis.

2. The paired internal iliac arteries are the principal arteries in the pelvis supplying the:
    a. Pelvic viscera
    b. Musculoskeletal pelvis
    c. Gluteal region
    d. Medial thigh
    e. Perineum

B. Branches of the internal iliac artery

1. Anterior and posterior branches
    a. Anterior generally supplies organs
    b. Posterior generally supplies walls and muscles
    c. Lots of variation
    d. ***I don't care where it comes from I only care where it goes***

2. Posterior division:
    a. Iliolumbar artery
        i. Ascends to lumbar region
    b. Lateral sacral arteries
        i. Supplies lateral portions of sacrum
    c. Superior gluteal artery
        i. Leaves to gluteal region between LS trunk and S1 ventral root

3. Anterior division:
    a. Umbilical artery (its occluded distal part continues as medical umbilical ligament)
    b. Superior vesical artery
    c. Inferior vesical artery
    d. Middle rectal artery
    e. Uterine artery (closely related to ureter)
    f. Vaginal artery
    g. Obturator artery (anomalous form: related to femoral hernia and may be lacerated during surgical repair of hernia)
    h. Internal pudendal artery
    i. Inferior gluteal artery

4. Posterior division:
    a. Iliolumbar artery
    b. Lateral sacral arteries
    c. Superior gluteal artery

C. Pelvic Veins

1. Pelvic venous plexuses: rectal, vesical, prostatic, uterine, and vaginal plexuses
    a. Tributaries of the internal iliac veins
    b. Internal iliac veins drain to common iliac veins then to inferior vena cava.

2. Venous anastomoses:
    a. Middle and Inferior rectal veins with superior rectal vein (caval portal anastomosis)
    b. Lateral sacral veins to internal vertebral venous plexus
        i. Pelvic cancer (prostatic or ovarian) may metastasis to vertebral and/or cranial sites.
    c. Branches of uterine veins to ovarian veins

D. Pelvic Nerves

1. Somatic nerves
    a. Obturator nerve: arising from L2-L4, supplying medical thigh, may be involved in ovarian disease, or injured during pelvic surgery
    b. Lumbosacral trunk: L4 and L5 joining sacral plexus
    c. Sacral plexus:
        i. Sciatic nerve: arising from L4-S3, innervating lower limb
        ii. Pudendal nerve: arising from S2-S4, innervating perineum
        iii. Superior gluteal nerve: arising from L4-S1, innervating gluteal muscles
        iv. Inferior gluteal nerve: arising from L5-S2, innervating gluteal muscles
        v. Posterior cutaneous nerve of thigh: arising from S1-S3, supplying skin over the posterior thigh.
        vi. Many small nerves to: piriformis, obturator internus, quadratus femoris, coccygeus and levator ani.

2. Pelvic autonomic innervation
    a. Parasympathetic fiber: via pelvic splanchnic nerves
        i. Arising from S2-S4
        ii. Hypogastric nerves and pelvic splanchnic nerves blend to form inferior hypogastric plexuses on either side of the pelvis.
    b. Sympathetic fibers: via superior hypogastric plexus
        i. Superior hypogastric plexus: continuation of inferior mesenteric and intermesenteric plexuses at the bifurcation of aorta
        ii. Hypogastric nerves connect superior and inferior hypogastric plexuses.
    c. Sacral sympathetic trunks
        i. Small branches to inferior hypogastric plexus
        ii. Primary function: vasomotor, pilomotor & sudomotor innervation of lower limb
    d. Visceral afferent fibers (VAF)
        i. Pelvic pain line: inferior limit of the peritoneum
        ii. Pain fibers above the pelvic pain line travel with sympathetic fibers retrograde to inferior thoracic/superior lumbar spinal ganglia.

iii. Pain fibers below the pelvic pain line follow parasympathetic fibers retrograde to spinal ganglia S2-S4 via pelvic splanchnic nerves.

iv. Unconscious reflexive fibers travel with parasympathetic fibers to S2-S4 spinal ganglia via pelvic splanchnic nerves

E. Lymph Drainage

1. Pelvic lymph node groups
   a. Sacral
   b. External iliac
   c. Internal iliac
   d. Common iliac
   e. Gonads to Lumbar/aortic

2. Perineal lymph node groups
   a. External genitalia goes to deep and superficial inguinal and internal iliac
   b. Gonads to Lumbar/aortic

Pelvic Viscera

1. Pelvic viscera includes the distal part of urinary system and gastrointestinal tract and organs of the reproductive systems.

F. Rectum

1. Continuous proximally with sigmoid colon at the rectosigmoid junction and distally with the anal canal at the anorectal junction.

2. S-shape at lateral view
   a. Sacral flexure
   b. Anorectal flexure (80°), important for fecal continence

3. Lateral flexures at anterior view
    a. Superior and inferior on the left
    b. Intermediate on the right
    c. Transverse rectal folds

4. Ampulla of the rectum, the distal dilated part accommodating fecal material

5. Arterial supply of the rectum
    a. Superior rectal artery:
    b. Middle rectal arteries
    c. Inferior rectal arteries
    d. Anastomoses

6. Venous drainage from plexuses under the mucous membrane
    a. Superior rectal vein
    b. Middle rectal veins
    c. Inferior rectal veins
    d. Portocaval anastomoses: collateral circulation in case of obstruction of portal vein

G. Anal Canal

1. Terminal part of the GI tract
2. Extending from where the ampulla narrows to the anus
3. Internal anal sphincter
    a. Thickening of the circular smooth muscle
    b. Tunic contraction at sympathetic stimulation
    c. Relaxation temporally at parasympathetic stimulation

4. External anal sphincter
    a. Voluntary striated muscle
    b. Innervated by inferior rectal nerve

5. Internal feature of the anal canal
    a. Anal columns: longitudinal ridges containing terminal branches of superior rectal artery
    b. Pectinate line (anal pecten, or white zone): dividing the anal canal into
        i. Superior part: visceral, derived from embryonic hindgut
        ii. Inferior part: somatic, derived from embryonic proctodeum

Arterial supply, venous and lymphatic drainage, and innervation of anal canal are summarized in the table below.

	Superior Anal Canal	**Inferior Anal Canal**
Arterial supply*	Superior rectal artery	Inferior rectal artery
Venous drainage*	Superior rectal vein to portal system	Inferior rectal vein to caval system
Lymphatic drainage	Internal iliac nodes	Superficial inguinal nodes
Sensory Innervation	Visceral; via inferior hypogastric plexuses Only sensitive to stretching	Somatic; via inferior rectal nerve. Sensitive to pain, touch, temperature

***Middle rectal arteries and veins provide anastomoses between superior and inferior rectal arteries and veins.**

6. Venous plexuses in the anal canal:
    a. Internal rectal venous plexus: deep to the mucosa of the anorectal junction
    b. External rectal venous plexus: external to the muscular wall of the anal canal

7. Lymphatic draining of the rectum and anus have no sex differences
    a. Distal anus to superficial inguinal
    b. Distal rectum to internal iliac
    c. Proximal rectal to sacral and pararectal

8. Internal hemorrhoids
    a. Prolapse of the rectal mucosa containing the veins of internal rectal venous plexus

9. External hemorrhoids
    a. Thromboses in the veins of the external rectal venous plexus

H. Ureters

1. Retroperitoneal muscular tube connecting the kidneys and the urinary bladder
    a. Passing over the pelvic brim/bifurcation of common iliac artery and entering pelvis
    b. Posterolateral to vas deferens (when present) and inferior to uterine artery (when present)
    c. Obliquely passing through the muscular wall of bladder to prevent reflux of urine
    d. Opening to ureteric orifices

2. Three restricted areas along the length of the ureters, which are the most common lodging sites for ureteric calculus:
    a. At the junction of the ureters and renal pelves
    b. As the ureters cross the pelvic brim
    c. Their passageway through the wall of the bladder

3. Arterial supply of ureters
    a. Numerous adjacent arteries supply the ureters along its length. Ureteric branches approach the ureters from medial to lateral in abdomen and from lateral to medial in pelvis. The ureters should be retracted accordingly during surgery to avoid interrupting their blood supply.

I. Urinary bladder

1. A hollow organ with strong muscular wall, as a temporary reservoir for urine
    a. Apex
    b. Fundus
    c. Body
    d. Neck

2. Superior surface is covered by peritoneum.
    a. When empty in adult, situated in lesser pelvis superoposterior to the pubic bones
    b. Retropubic space of Retzius allows extension.
    c. Its neck is firmly held by condensations of the pelvic fascia:
        i. Lateral ligaments of bladder
        ii. Pubovesical ligament in the absence of a prostate gland
        iii. Puboprostatic ligament when prostate gland is present
        iv. Tendinous arch of the pelvic fascia

3. Internal features of the bladder:
    a. Ureteric orifices
    b. Trigone of the bladder
    c. Uvula of bladder
    d. Detrusor muscle
    e. Internal urethral sphincter, present when needed to prevent ejaculatory reflux of semen into bladder

4. Arterial supply of the bladder
    a. Superior vesical arteries
    b. Inferior vesical arteries
    c. Vaginal arteries replace the inferior vesical arteries when the vagina is present.

5. Venous drainage of the bladder
    a. Vesical venous plexus
    b. Prostatic venous plexus (when prostate is present)
    c. Internal iliac veins
    d. May drain to internal vertebral venous plexuses via sacral veins.

6. Innervation of the bladder
    a. Vesical plexuses from inferior hypogastric plexuses
    b. Sympathetic fibers: stimulate contraction of internal urethral sphincter at ejaculation to prevent reflux of semen into bladder during ejaculation of sperm
    c. Parasympathetic fibers: motor to detrusor muscle and inhibitory to internal urethral sphincter in all individuals
    d. Sensory fibers:
        i. Above pain line refers to T11-L2ish
        ii. Below pain line to S2-S4

J. Urethra

1. Sex differences
2. Estrogen driven
    a. Short
    b. Opens to vaginal vestibule
    c. No defined internal sphincter

3. Testosterone drives formation of a long muscular tube from: with 4 parts:
    e. Intramural (preprostatic) part
    f. Prostatic urethra: widest and most distensible
    g. Intermediate (membranous) urethra: shortest and least distensible, penetrating through the deep perineal pouch
    h. Spongy urethra: 15-16cm long runs in corpus spongiosum
4. Blood supply of the bladder urethra
    a. branches from the internal iliac
        i. Superior and inferior vesicle
    b. Perineal portions of the urethra supplied by the internal pudenal branches
5. Lymphatic draining of the rectum and anus have no sex differences
    a. Bladder to internal and external iliac
    b. Distal urethra superficial inguinal
    c. Prostatic and spongy urethra is mixed

Answers to Study Questions

These answers are not meant to be all-encompassing; rather they may be examples of the types of thought processes needed for application of the material.

1. A44-year-old patient presents with rectal bleeding. She describes the blood as bright red. She does not experience pain in the anal region. Further examination reveals internal hemorrhoids. You recall that internal hemorrhoids stimulate sensory fibers of which of the following nerves?

   A. Inferior gluteal
   B. Inferior hypogastric plexus
   C. Inferior rectal
   D. Obturator
   E. Posterior femoral cutaneous

Answer: The internal rectal venous plexus is in the submucosa in the anorectal junction and anal canal. It receives abundant venous anastomoses from superior, middle, and inferior rectal veins, and arteriovenous anastomoses from superior and middle rectal arteries. Breakdown of the muscularis mucosa or portal hypertension can cause prolapse of the rectal mucosa. This condition is referred as internal hemorrhoid. Internal hemorrhoids can prolapse through the anal canal and are compressed by the anal sphincters. As a result, they can be strangulated and ulcerated. Because the abundant arteriovenous anastomoses present in the submucosa, bleeding from internal hemorrhoids is characteristically bright red. Internal hemorrhoids are above the pectinate line and receive visceral sensory nerve fibers from the inferior hypogastric plexuses. Visceral afferent fibers are sensitive to stretch but not to pain and temperature, thus, internal hemorrhoids are not painful and can be treated without anesthesia.

Based on this question, students should explore further the external hemorrhoids, the innervation, arterial supply, venous and lymphatic drainage of the anal canal superior or inferior to the pectinate line.

2. A 42-year-old patient was diagnosed with ovarian cancer. The cancerous ovary was surgically removed.

   Question A: During the surgery, you recall which nerve is vulnerable to injury?

   Obturator nerve. Obturator nerve originates from L2-L4 in the posterior abdominal wall. It enters the pelvis and runs along the lateral wall of the pelvis to reach the obturator canal. It passes through the canal to innervate the medial thigh.

   In the pelvis, the obturator nerve is closely related to the ovarian fossa and ovary. Surgeons need to pay special attention to this relationship during surgical removal of diseased ovary.

   Question B: During the surgery, you need to ligate the ovarian vessels. Which structure might be at risk during this procedure?

   Ureter. Ureters run inferiorly along the posterior abdominal wall. At the level of pelvic brim, ureters are closely associated with the ovarian vessels. Ligation of the ovarian vessel could put the ureter at risk.

3. Based on the branches of the internal iliac artery, which body regions receive arterial supply via internal iliac artery?

   a. Musculoskeletal pelvis--iliolumbar and lateral sacral arteries
   b. Pelvic viscera--superior/inferior vesical, middle rectal, uterine, and vaginal arteries
   c. Perineum--internal pudendal artery
   d. Gluteal region--superior/inferior gluteal arteries
   e. Medial thigh--obturator artery

4. At which structures can we see the urinary system and reproductive system merge?

Prostatic urethra. Ductus deferens joins the duct of the seminal glands to form the ejaculate duct at the posterior/inferior aspect of the urinary bladder. The ejaculatory duct penetrates the prostate and opens onto the seminal colliculus of the prostatic urethra.

5. You are performing a rectal examination on your 72-year-old patient. Based on the pelvic anatomy you learned at medical school, recall what structures you can palpate in a patient.

Digital rectal examination is a common procedure to examine the pelvic viscera. Structures anterior to the inferior part of the rectum can be palpated through its wall. These will include the prostate and seminal glands, or the uterine cervix. Posteriorly, the anterior surface of the lower sacrum and coccyx, the ischial spines and tuberosities are also palpable. Enlarged internal iliac lymph nodes, pathological thickening of ureters, swelling of ischioanal fossae may be palpated.